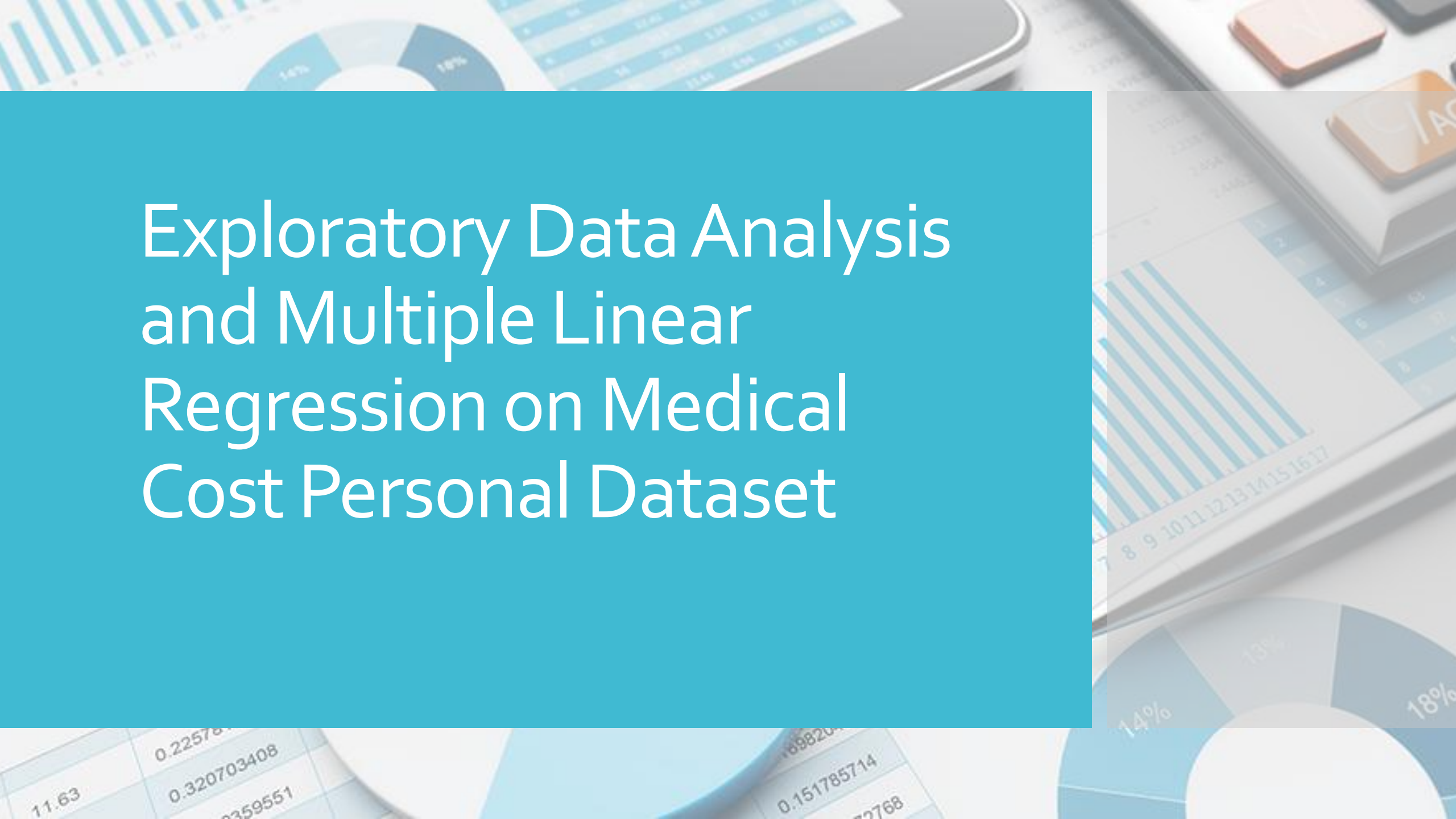


The background of the slide features a collage of financial and data-related imagery. It includes a close-up of a calculator with orange and grey buttons, several bar charts with blue bars on white backgrounds, and pie charts with segments in shades of blue and grey. Some pie chart segments are labeled with percentages like 14%, 13%, and 18%. The overall theme is data analysis and finance.

Exploratory Data Analysis and Multiple Linear Regression on Medical Cost Personal Dataset

Abstract

- **Overview:** Analysis of Kaggle's Medical Cost Personal Dataset.
- **Objective:** Explore factors affecting medical costs (age, BMI, smoking).
- **Key Findings:**
 - Age, smoking status, and obesity significantly affect medical charges.
 - Final regression model explains 84.15% of variance.

Introduction and Objectives

Objective 1:

- Explore relationships between demographics and medical costs.

Objective 2:

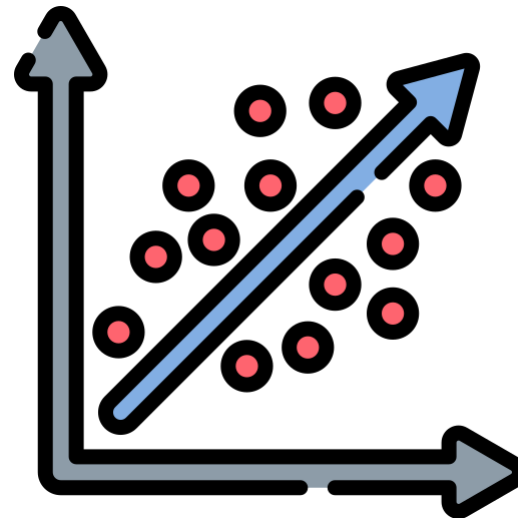
- Evaluate predictive power of multiple regression models.

Objective 3:

- Investigate the interaction between smoking and obesity as key factors.

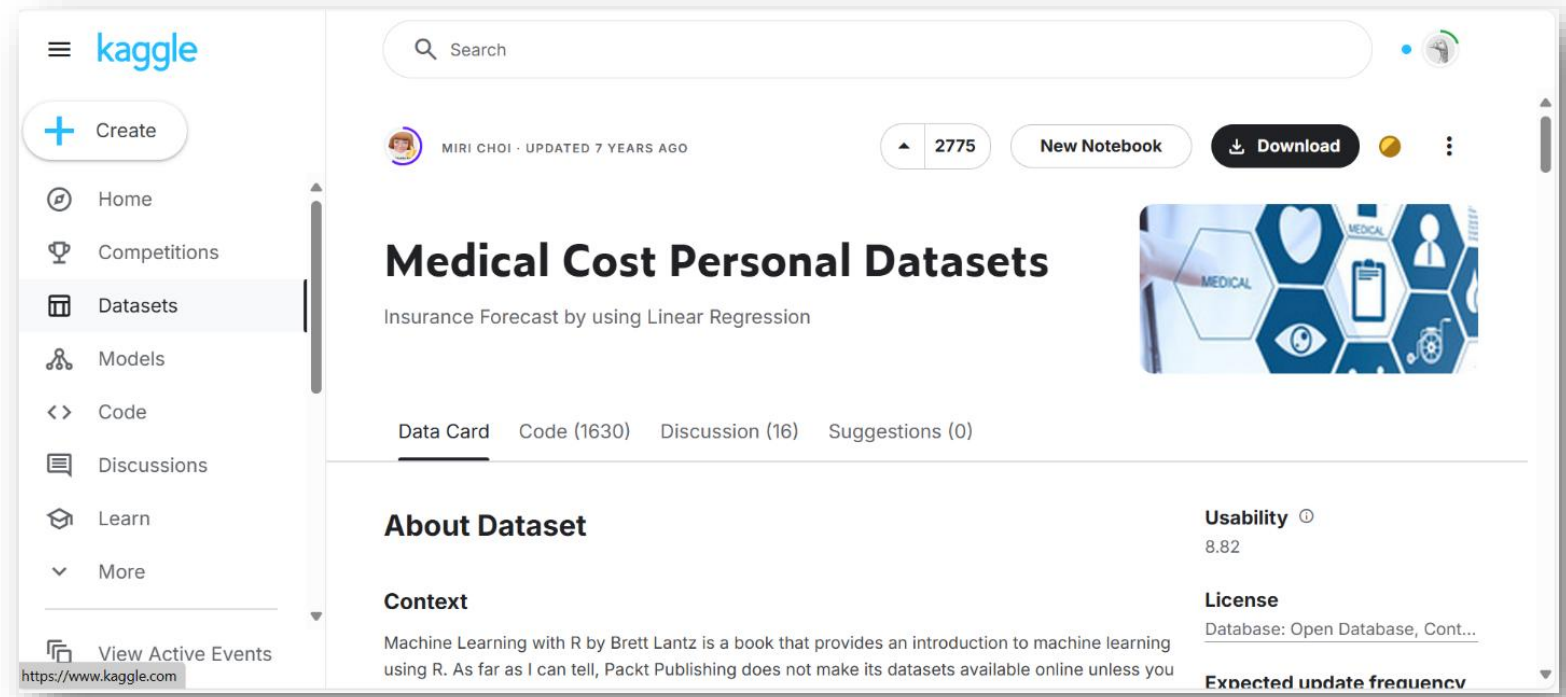
Background/ Literature Review

- Regression analysis is used to understand relationships between variables.
- Previous studies emphasize smoking and obesity's effect on medical costs.
- The interaction between smoking and obesity is significant in influencing health costs.

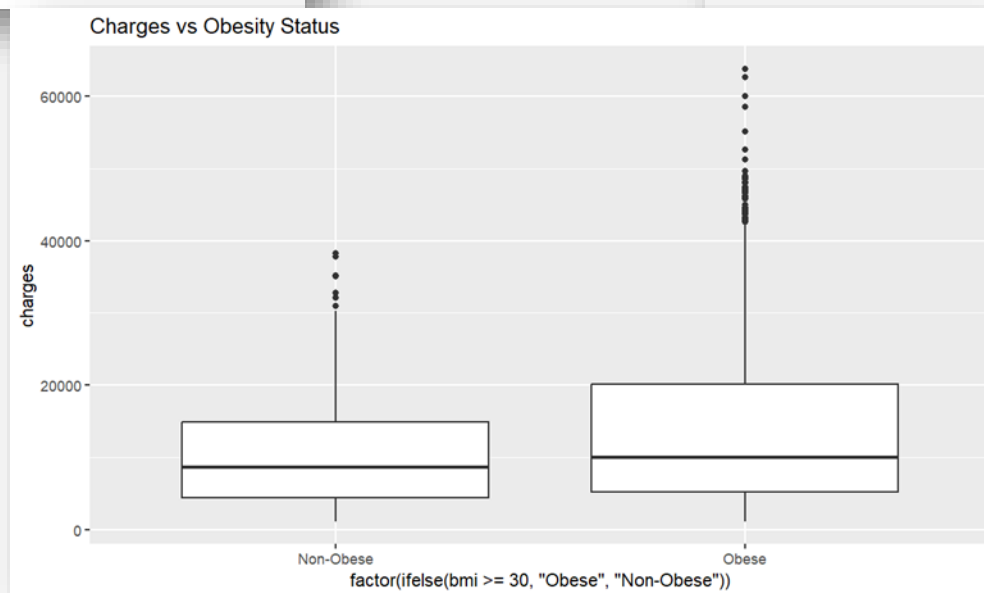
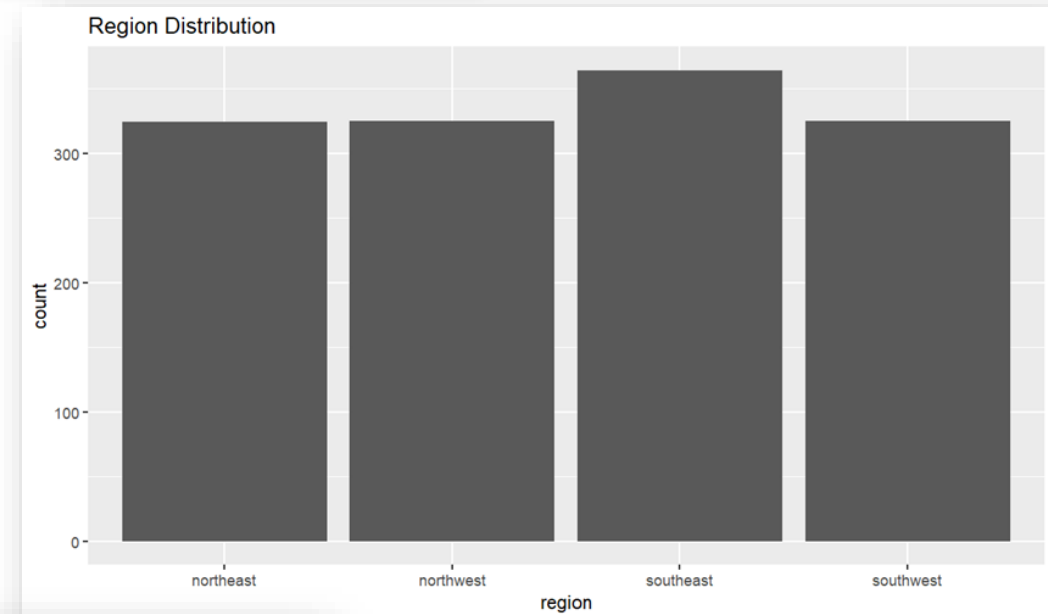
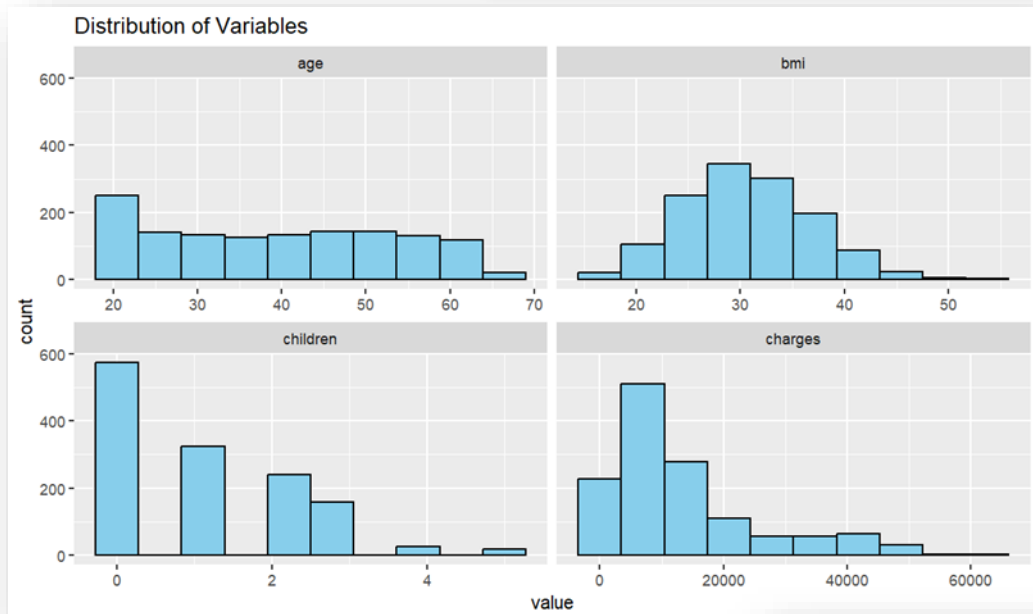


Data Description

- **Dataset Source:** Kaggle – Medical Cost Personal Dataset.
- **Key Variables:**
 - Dependent: Charges (Insurance cost).
 - Independent: Age, BMI, Smoking status, Region, and Children.
 - Additional Feature: Obese (BMI = 30) to capture obesity's direct impact.



Exploratory Data Analysis (EDA)



Multiple Linear Regression (Model 2)

```
> lm.fit.m.2 <- lm(charges ~ age + children + smoker + obese, data = train)
> summary(lm.fit.m.2)
```

```
Call:
lm(formula = charges ~ age + children + smoker + obese, data = train)
```

```
Residuals:
    Min       1Q   Median       3Q      Max
-12729.8  -3664.1    81.5   1375.4  26733.1
```

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -4666.99     613.31  -7.609 6.35e-14
age           264.14       13.54  19.509 < 2e-16
children      439.11       158.22   2.775 0.00562
smokeryes     23571.85     469.60  50.196 < 2e-16
obeseyes       4305.81      382.37  11.261 < 2e-16
```

```
(Intercept) ***
age          ***
children     **
smokeryes    ***
obeseyes     ***
---
Signif. codes:
0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 6017 on 998 degrees of freedom
Multiple R-squared:  0.7531,    Adjusted R-squared:  0.7521
F-statistic: 760.8 on 4 and 998 DF,  p-value: < 2.2e-16
```

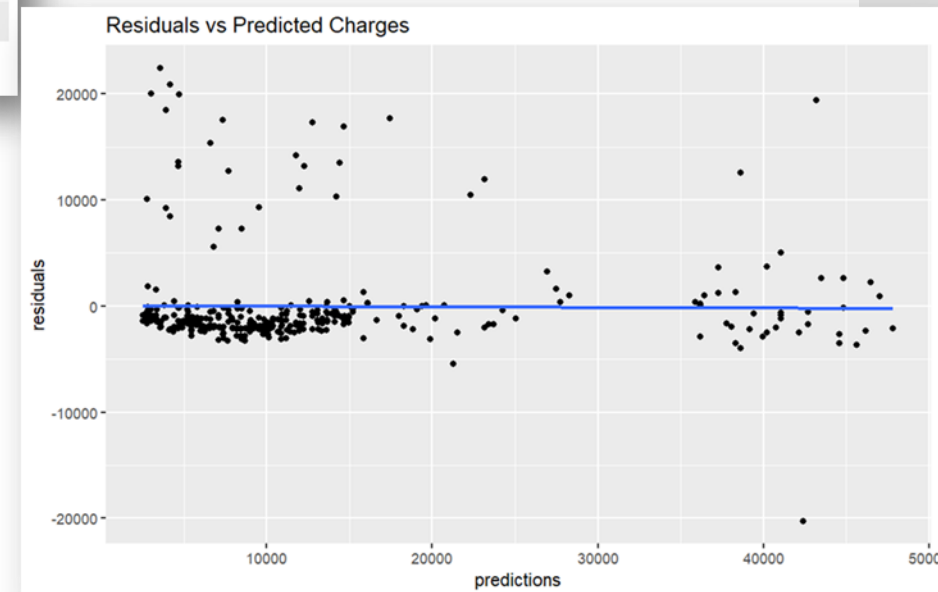
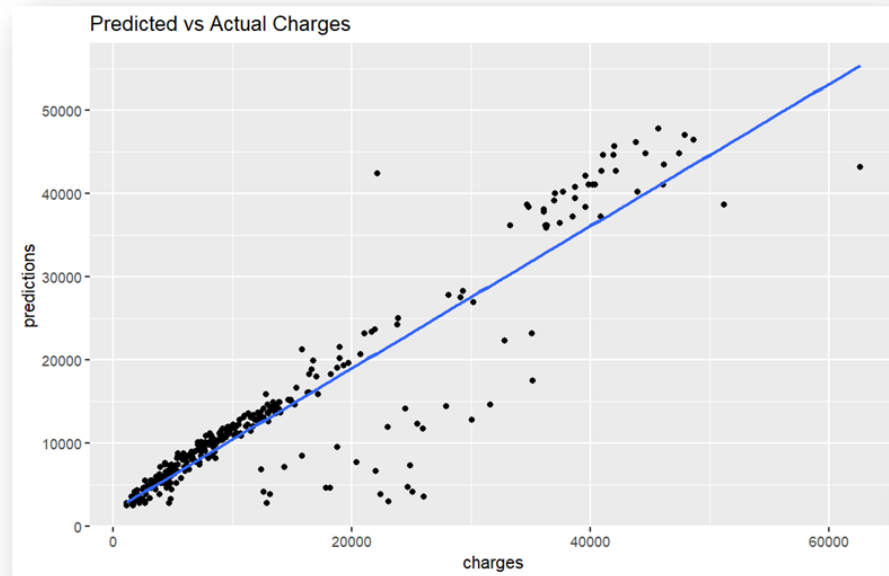
- **Variables Included:** Age, Children, Smoker, Obese.
- **Justification:** These variables directly influence healthcare costs.
- **Adjusted R-Squared:** 0.7521 (explains 75.21% of variability).

Regression Results (Model 2)

- **Significant Predictors:** Age, Smoker, Obese (p-values < 0.05).
- **Residuals Check:** Improved distribution and variance compared to Model 1.

Model Assumptions

- Residual plots indicate good fit.
- Check for multicollinearity and further refinement needed.



Model Comparison

Model	Formula	Adjusted R-squared	Significance	Residual Standard Error	Interpretation
Model 1	charges ~ age + bmi + sex + children + smoker + region	0.7521	All predictors significant except sex and region.	6017	Reasonable fit but includes insignificant predictors; simplifying improves performance.
Model 2	charges ~ age + children + smoker + obese	0.7521	Significant predictors: age, smoker, obesity.	6017	Similar performance to Model 1 but more interpretable by removing irrelevant predictors.
Model 3	charges ~ obese * smoker + age + children	0.8635	Interaction between smoker and obesity significant.	4464	Significant improvement in fit, highlighting the interaction between smoking and obesity.
Model 4	charges ~ obese * smoker + age	0.8610	Significant interaction, age also significant.	4506	Simplified version of Model 3, balancing simplicity and accuracy.

- Model 3 (Adjusted R-Squared: 0.8635) provides the best fit with interaction terms.

Hypothesis Testing

- **Null Hypothesis (H_0):** Predictor has no significant relationship with charges.
- **Key Results:**
 - Age: Significant ($p < 0.001$).
 - Smoker: Significant ($p < 0.001$).
 - Obesity alone: Not significant.
 - Interaction between smoking and obesity: Significant ($p < 0.001$).

Contextual Insights

Insights for Insurers and Policymakers:

- Age, smoking, and obesity interact to significantly impact insurance charges.
- Smoking cessation and obesity reduction strategies could reduce healthcare costs.

Discussion/ Analysis

- Model 1 had a good fit but included non-significant variables.
- Model 2 removed irrelevant variables, improving interpretability.
- Model 3 added interactions between smoking and obesity, improving fit.

Conclusion

- **Key Findings:** Age, smoking, and obesity (interaction) are key predictors.
- **Best Model:** Model 3 (Adjusted R-Squared = 0.8635).
- **Recommendations:**
 - **Insurers:** Adjust premiums based on key predictors.
 - **Policymakers:** Focus on smoking and obesity reduction.
 - **Future Research:** Explore additional variables and non-linear models.

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